Lecture 3 — part 2 Conformal Prédiction

vendredi 6 février 2026 13:27

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Conformal Prediction – some insights

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February 6, 2026

Lecture 3, M2 - Maths/Info & PSND

General Framework

Supervised learning setting:
- Training data $(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \dots, (\mathbf{X}_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$
  - Regression: $\mathcal{Y} \subset \mathbb{R}$
  - Classification: $\mathcal{Y} = [K]$ with $K \in \mathbb{N}^*$

Goals:
- Inference on the parameters of the regression
  - Distribution of Y given $\mathbf{X}$
  - Estimation / Tests / Confidence intervals
  - May require some hypothesis on the model
    - Parametric (linear regression) / Non-parametric (smoothness) / ...
- Predictive inference
  - Predict value of Y given $\mathbf{X}$ for test points $\mathbf{X}_{n+1}, \mathbf{X}_{n+2}, \dots$
  - Build prediction intervals for Y given $\mathbf{X}$ for test points $\mathbf{X}_{n+1}, \mathbf{X}_{n+2}, \dots$

$Y = X\beta^* + \varepsilon$

$\mathcal{D}_n = \{(\dots)\}$ Training set $\rightsquigarrow \boxed{\hat\beta}$

- Estimation : Is $\hat\beta$ good to estimate β^*
  $\|\hat\beta - \beta^*\|_2 < \dots \to 0$?
- Test : $\beta_j^* = 0$ for s
- CI : $\beta_j^* \in [\hat\beta_j \pm \delta]$
- Prediction : goal est $X^T\beta^*$

$\mathbb{E}\left[(X_{n+1}^T\beta^* - X_{n+1}^T\hat\beta)^2\right]$

$X_{n+1}^T\beta^* \in [X_{n+1}^T\hat\beta \pm \partial]$

Focus on prediction intervals

Build prediction sets / intervals $\Gamma : \mathcal{X} \to 2^{\mathcal{Y}}$ with high coverage guarantees: for $\alpha \in (0,1)$
- Average[1] $\mathbb{P}(Y \in \Gamma(\mathbf{X})) \geq 1 - \alpha$
- Instance-conditional $\mathbb{P}(Y \in \Gamma(\mathbf{X}) | \mathbf{X} = \mathbf{x}) \geq 1 - \alpha$, for $\mathbf{x} \in \mathcal{X}$
- Label-conditional $\mathbb{P}(Y \in \Gamma(\mathbf{X}) | Y = y) \geq 1 - \alpha$, for $y \in \mathcal{Y}$

Construct Γ based on training sample $\mathcal{D}_n = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$
- That is $\hat\Gamma(\cdot) = \hat\Gamma_n(\cdot, \mathcal{D}_n)$

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[1] $(\mathbf{X}, Y) \sim \mathbb{P}$

Prediction intervals – new style

Build prediction sets / intervals $\Gamma : \mathcal{X} \to 2^{\mathcal{Y}}$ with high *finite-sample* coverage guarantees

"Old" fashion: rely on model specification
- (Non)-parametric assumptions on the regression function
- Use classical tools for estimation, *e.g.*, least-squares, kernels,...
- $Y_{n+1} \in \Gamma(\mathbf{X}_{n+1})$ based on the previous estimation

"New" fashion: **conformal prediction**
- Make as less assumptions as possible
- Only one requirement: $\Gamma(\mathbf{X}_{n+1})$ attains the prescribed coverage
- Very flexible: it can be applied to any machine learning model

Introduction for conformal prediction

What do we need to build a prediction interval in Conformal prediction style:
- Level of coverage $1 - \alpha$ (significance level $\alpha \in (0,1)$)
- Measure of (non) conformity $s(\mathbf{x}, y)$ based on some prediction function $g : \mathcal{X} \mapsto \mathcal{Y}$ *e.g.*, $s(\mathbf{x}, y) = |y - g(\mathbf{x})|$

That's it!

Pre-requirements: technical tools

Technical consideration 1: Exchangeability

Can we have universal validity?
- Not possible to cover every joint distribution on the data....

Definition

Random variables $Z_1, \dots, Z_n$ are exchangeable if

$$(Z_1, \dots, Z_n) \overset{d}{=} (Z_{\sigma(1)}, \dots, Z_{\sigma(n)})$$

for every permutation $\sigma \in \mathcal{S}_n$.

- An important special case:
  if $Z_1, \dots, Z_n \overset{iid}{\sim} P$ then $Z_1, \dots, Z_n$ are exchangeable
- Exchangeable but not i.i.d.: An equicorrelated model

$$\begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & \rho & \dots & \rho \\ \rho & 1 & \dots & \rho \\ \dots & \dots & \dots & \dots \\ \rho & \rho & \dots & 1 \end{pmatrix}\right)$$

Technical consideration 2: CDF and quantiles

Definition
- Empirical cumulative distribution functions (emp. CDF)

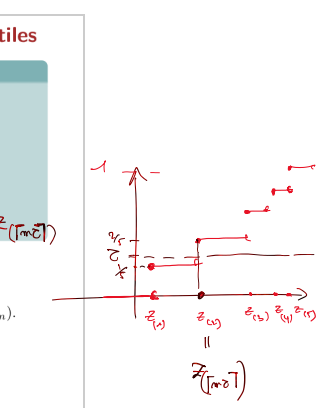
$$\hat F_m(t) = \frac{1}{m}\sum_{i=1}^m \mathbf{1}_{\{Z_i \leq t\}}$$

- The empirical τ-quantile of the list $Z_1, \dots, Z_m \in \mathbb{R}$ is

$$\hat Q_m^\tau(Z_1, \dots, Z_m) = \inf\{t \in \mathbb{R} : \hat F_m(t) \geq \tau\} = Z_{(\lceil m\tau \rceil)}$$

Main results:
1. Let $\tau \in [0,1]$. For any $z_1, \dots, z_m, z \in \mathbb{R}$,
   $$z \leq \hat Q_{m+1}^\tau(z_1, \dots, z_m, z) \iff z \leq \hat Q_m^{\tau(1+1/m)}(z_1, \dots, z_m).$$
2. Let $\tau \in [0,1]$. If $Z_1, \dots, Z_m \in \mathbb{R}$ are exchangeable
   $$\tau \leq \mathbb{P}\left(Z_i \leq \hat Q_m^\tau(Z_1, \dots, Z_m)\right) \underset{\text{if } Z_i \text{ cont.}}{\leq} \tau + \frac{1}{m}$$

exact



Full conformal prediction $\rightsquigarrow$ the regression case!

First attempt

- Assume data $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ are exchangeable
- Let $g : \mathcal{X} \mapsto \mathcal{Y}$ $\rightsquigarrow$ score

Compute for (small) $\alpha \in (0,1)$ $\boxed{(1-\alpha) = \tau}$

$$q^{(1-\alpha)^+} = \hat Q_n^{(1-\alpha)(1+\frac{1}{n})}\left(\{|Y_i - g(\mathbf{X}_i)|\}_{i=1}^n\right)$$

$s(x,y) = |y - g(x)|$

$\mathcal{D}_n$ $\underbrace{|Y_i - g(X_i)| \dots}_{Z_i}$

Define the prediction intervals for test point $\mathbf{X}_{n+1}$

$$\Gamma(\mathbf{X}_{n+1}) = \left[g(\mathbf{X}_{n+1}) \pm q^{(1-\alpha)^+}\right]$$

$Z_{n+1} = |Y_{n+1} - g(X_{n+1})|$

Then[2]

$\mathbb{P}(Y_{n+1} \in \Gamma(\mathbf{X}_{n+1})) =$

$\mathbb{P}\left(|Y_{n+1} - g(\mathbf{X}_{n+1})| \leq \hat Q_{n+1}^{1-\alpha}\left(\{|Y_i - g(\mathbf{X}_i)|\}_{i=1}^{n+1}\right)\right) \geq 1 - \alpha$

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[2] Due to the fact that exchangeability of $(\mathbf{X}_i, Y_i)$'s implies exchangeability of the $|Y_i - g(\mathbf{X}_i)|$'s

Using the training set for inference

BUT what is g – that is used to build the score s?
- It should be trained on $\mathcal{D}_n$ based on some algorithm $\mathcal{A}$?
  $\hookrightarrow g(\cdot) \rightsquigarrow \hat g_n(\cdot, \mathcal{D}_n) = \mathcal{A}(\mathcal{D}_n)$
- This may induce over-fitting and then

$$\underbrace{|Y_{n+1} - \hat g_n(\mathbf{X}_{n+1})|}_{\text{test error}} \gg \underbrace{\frac{1}{n}\sum_{i=1}^n |Y_i - \hat g_n(\mathbf{X}_i)|}_{\text{avg. training error}}$$

Generalization error

in-sample error

even if training & test data are from the same distribution

$\hat g$ is built on $\mathcal{D}_n$

$\hat g$ known $\mathcal{D}_n$ but not (X_{n+1}, Y_{n+1})

$\hat Q^\alpha$ is $|Y_1 - \hat g_n(X_1)|, \dots, |Y_n - \hat g_n(X_n)|, |Y_{n+1} - \hat g_n(X_{n+1})|$ exchangeable

NO!

Full conformal prediction: preliminaries

Key ideas:
- Add artificially a label to $\mathbf{X}_{n+1} \rightsquigarrow \bar y \in \mathcal{Y}$
- Consider the extended dataset $((\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, \bar y)) := \boxed{\mathcal{D}_{n+1}^{\bar y}}$ for $\bar y$
- Build scores based on these samples:
  - Train the prediction function: $\hat g^{\bar y}(\cdot) = \mathcal{A}(\mathcal{D}_{n+1}^{\bar y})$
  - Compute scores
    $S_i^{\bar y} = s(\mathbf{X}_i, Y_i) = |\hat g^{\bar y}(\mathbf{X}_i) - Y_i|$ $\quad S_{n+1}^{\bar y} = s(\mathbf{X}_{n+1}, \bar y) = |\hat g^{\bar y}(\mathbf{X}_{n+1}) - \bar y|$
- If $\mathcal{A}$ is **symmetric**, $\forall Z_1, \dots, Z_m$ and $\forall \sigma \in \mathcal{S}_m$ permutation $\mathcal{A}(Z_1, \dots, Z_m) = \mathcal{A}(Z_{\sigma(1)}, \dots, Z_{\sigma(m)})$ the exchangeability of $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ implies exchangeability of $S_1^{Y_{n+1}}, \dots, S_n^{Y_{n+1}}, S_{n+1}^{Y_{n+1}}$

Full conformal prediction: theory

Full conformal prediction for test point $\mathbf{X}_{n+1}$ is defined as

$$\hat\Gamma(\mathbf{X}_{n+1}) = \{\bar y \in \mathcal{Y} : S_{n+1}^{\bar y} \leq \hat Q_n^{(1-\alpha)(1+1/n)}(S_1^{\bar y}, \dots, S_n^{\bar y})\}$$

Theorem (Full conformal[3])

[3] Vovk et al 2005, Algorithmic Learning in a Random World

If data in $\mathcal{D}_{n+1}^{Y_{n+1}}$ are exchangeable and $\mathcal{A}$ is symmetric, the full conformal prediction satisfies

$$\mathbb{P}\left(Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1})\right) \geq 1 - \alpha$$

Proof

$$\begin{aligned} Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1}) &\iff S_{n+1}^{Y_{n+1}} \leq \hat Q_n^{(1-\alpha)(1+1/n)}(S_1^{Y_{n+1}}, \dots, S_n^{Y_{n+1}}) \\ &\iff S_{n+1}^{Y_{n+1}} \leq \hat Q_{n+1}^{1-\alpha}(S_1^{Y_{n+1}}, \dots, S_{n+1}^{Y_{n+1}}) \end{aligned}$$

- This holds with prob. $\geq 1 - \alpha$

$Y_1 - \hat g^{Y_{n+1}}(X_1), \dots, Y_{n+1} - \hat g^{Y_{n+1}}(X_{n+1})$ are exchangeable.

~~Conformal prediction = outlier detection~~

~~Other interpretation:~~

~~Is Z_{n+1} exchangeable with $Z_1, \dots, Z_n$?~~

~~$\Updownarrow$~~

~~Is Z_{n+1} an inlier relative to $Z_1, \dots, Z_n$?~~

~~Conformal prediction = outliers detection~~

~~We can reinterpret full Conformal prediction as a hypothesis test~~
- ~~For each $\bar y \in \mathcal{Y}$ define the null hypothesis~~

$$H_0(\bar y) : Y_{n+1} = \bar y$$

- ~~Let $\hat\Gamma(\mathbf{X}_{n+1}) = \{\bar y \in \mathcal{Y} : H_0(\bar y)$ is not rejected at level $\alpha\}$~~

~~**The p-value for testing** $H_0(\bar y)$~~

$$p(\bar y) = \frac{1 + \sum_{i=1}^n \mathbf{1}_{S_i^{\bar y} \geq S_{n+1}^{\bar y}}}{n+1}$$

~~That is, $\hat\Gamma(\mathbf{X}_{n+1}) = \{\bar y \in \mathcal{Y} : p(\bar y) > \alpha\}$~~

~~Conformal prediction = outliers detection~~

~~**Theorem (validity of the "conformal" p-value)**~~

~~*If $Z_1, \dots, Z_{n+1}$ are exchangeable, then for all $\alpha \in [0,1]$*~~

$$\mathbb{P}\left(p(Z_{n+1}) \leq \alpha\right) \leq \alpha$$

~~*where*~~

$$p(Z_{n+1}) = \frac{1 + \sum_{i=1}^n \mathbf{1}_{Z_i \geq Z_{n+1}}}{n+1}$$

~~**Proof**~~

~~Exchangeability implies~~

$$\{p(Z_{n+1}) > \alpha\} = \left\{Z_{n+1} \leq \hat Q_{n+1}^{1-\alpha}(Z_1, \dots, Z_{n+1})\right\}$$

~~that holds with probability $\geq 1 - \alpha$~~

Full Conformal Prediction: drawback

Full conformal prediction for test point $\mathbf{X}_{n+1}$

$$\hat\Gamma(\mathbf{X}_{n+1}) = \left\{\bar y \in \mathcal{Y} : S_{n+1}^{\bar y} \leq \hat Q_n^{(1-\alpha)(1+1/n)}(S_1^{\bar y}, \dots, S_n^{\bar y})\right\}$$

- As previously said $\hat g^{\bar y}(\cdot) = \mathcal{A}(\mathcal{D}_{n+1}^{\bar y})$ with $\mathcal{D}_{n+1}^{\bar y} = \{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, \bar y)\}$
- For each test point $\mathbf{X}_{n+1}$, algorithm $\mathcal{A}$ is applied for all value $\bar y \in \mathcal{Y}$
  - Fine in classification with $|\mathcal{Y}| = L$ and L is small.
  - Problem in regression or in classification with large K!
- Full conformal prediction may be computationally not efficient!

Full conformal prediction: First solution

Trimmed conformal prediction[4]
- Truncate to the empirical range:

$$\hat\Gamma(\mathbf{X}_{n+1}) = \left\{\bar y \in \left[\min_{1\leq i\leq n} Y_i, \max_{1\leq i\leq n} Y_i\right] : S_{n+1}^{\bar y} \leq \hat Q_n^{(1-\alpha)(1+1/n)}(S_1^{\bar y}, \dots, S_n^{\bar y})\right\}$$

Then (coverage):

$$\begin{aligned} &\mathbb{P}(Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1})) \\ &= \mathbb{P}\left\{(s(\mathbf{X}_{n+1}, Y_{n+1}) \leq \hat q) \cap \left\{Y_{n+1} \in \left[\min_{1\leq i\leq n} Y_i, \max_{1\leq i\leq n} Y_i\right]\right\}\right\} \\ &\geq \mathbb{P}(s(\mathbf{X}_{n+1}, Y_{n+1}) \leq \hat q) - \mathbb{P}\left\{Y_{n+1} \notin \left[\min_{1\leq i\leq n} Y_i, \max_{1\leq i\leq n} Y_i\right]\right\} \\ &\geq 1 - \alpha - \frac{2}{n+1} \end{aligned}$$

- Loss in accuracy $(-\frac{2}{n+1}) \rightsquigarrow$ solvable!
- Discretization is still needed!

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[4] Chen et al. 2016, Trimmed conformal prediction for high-dimensional models.

Full conformal prediction: Second solution

Exploit (piecewise) linearity in $\bar Y$ of $\mathcal{A}$[4]
- Fitting data points has the form:

$$\begin{pmatrix} \hat g^{\bar y}(\mathbf{X}_1) \\ \vdots \\ \hat g^{\bar y}(\mathbf{X}_n) \\ \hat g^{\bar y}(\mathbf{X}_{n+1}) \end{pmatrix} = H \cdot \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \\ \bar y \end{pmatrix}$$

linear prediction

- $H = H(\mathbf{X}_1, \dots, \mathbf{X}_{n+1})$ independent of $Y_1, \dots, Y_n, \bar y$

Examples:
- Linear regression & ridge regression
- Kernel smoothing type methods
- Lasso (piecewise-linear)

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[4] Vovk et al 2005, Algorithmic Learning in a Random World

Full conformal prediction: Second solution

How to run full conformal prediction in this special case:[5]
1. Compute conformity scores vector

$$\begin{pmatrix} Y_1 \\ \vdots \\ Y_n \\ \bar y \end{pmatrix} - \begin{pmatrix} \hat g^{\bar y}(\mathbf{X}_1) \\ \vdots \\ \hat g^{\bar y}(\mathbf{X}_n) \\ \hat g^{\bar y}(\mathbf{X}_{n+1}) \end{pmatrix} = \underbrace{(I - H) \cdot \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \\ 0 \end{pmatrix}}_{=A} + \underbrace{(I - H) \cdot \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}}_{B} \bar y$$

2. Prediction interval can now be computed in finite time:

$$\hat\Gamma(\mathbf{X}_{n+1}) = \left\{\bar y : |\bar y \cdot (a_{n+1} + b_{n+1}\bar y)| \leq \hat Q_n^{(1-\alpha)(1+1/n)}\left(\{|Y_i - (a_i + b_i\bar y)|\}_{i=1}^n\right)\right\}$$

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[5] Burnaev & Vovk 2014, Efficiency of conformalized ridge regression

Computational solutions for full conformal prediction

- Restrict to a grid of y values
- Specialized methods for specific algorithms/settings, e.g., Ridge[6], Lasso[7], stable algorithms[8]
- Discretized CP[9] – use a discretized version of $\mathcal{A}$ to restore theoretical guarantees

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[6] Burnaev & Vovk 2014, Efficiency of conformalized ridge regression

[7] Hebiri 2010, Lasso conformal prediction

[8] Ndiaye 2022, Stable Conformal Prediction Sets

[9] Chen et al. 2017, Discretized conformal prediction for efficient distribution-free inference

Alternative approach: Split conformal prediction

Full conformal prediction on data!

Notice that $Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1})$ reads as

$$S_{n+1}^{Y_{n+1}} \leq \hat Q_{n+1}^{(1-\alpha)}(S_1^{Y_{n+1}}, \dots, S_{n+1}^{Y_{n+1}}).$$

Then, for "at least" $n(1-\alpha)$ indices i

$$|\hat g^{Y_{n+1}}(\mathbf{X}_{n+1}) - Y_{n+1}| \leq |\hat g^{Y_{n+1}}(\mathbf{X}_i) - Y_i|$$

Let $\hat q_{n+1} = \hat Q_{n+1}^{(1-\alpha)}\left(\{|\hat g^{Y_{n+1}}(\mathbf{X}_i) - Y_i|\}_{i=1}^{n+1}\right)$.

$$|\hat g^{Y_{n+1}}(\mathbf{X}_{n+1}) - Y_{n+1}| \leq \hat q_{n+1} \iff Y_{n+1} \in \left[\hat g^{Y_{n+1}}(\mathbf{X}_{n+1}) \pm \hat q_{n+1}\right]$$

If we forget about Y_{n+1}, we can suggest the alternative prediction set:

$$\tilde\Gamma(\mathbf{X}_{n+1}) = \left[\hat g(\mathbf{X}_{n+1}, \mathcal{D}_n) \pm \hat q_n^{(1-\alpha)^+}\right]$$

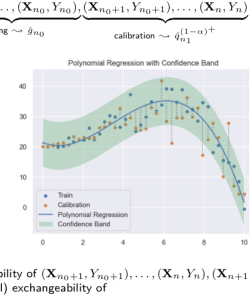
with $\hat q_n^{(1-\alpha)^+} = \hat Q_n^{(1-\alpha)(1+1/n)}\left(\{|\hat g(\mathbf{X}_i, \mathcal{D}_n) - Y_i|\}_{i=1}^n\right)$.

- *Exchangeability* fails!!

Split conformal predictor

The data: with $n_0 + n_1 = n$

$$\underbrace{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_{n_0}, Y_{n_0})}_{\text{training} \rightsquigarrow \hat g_{n_0}}, \underbrace{(\mathbf{X}_{n_0+1}, Y_{n_0+1}), \dots, (\mathbf{X}_n, Y_n)}_{\text{calibration} \rightsquigarrow \hat q_{n_1}^{(1-\alpha)^+}}, \underbrace{(\mathbf{X}_{n+1}, Y_{n+1})}_{\text{test}}$$



- Exchangeability of $(\mathbf{X}_{n_0+1}, Y_{n_0+1}), \dots, (\mathbf{X}_n, Y_n), (\mathbf{X}_{n+1}, Y_{n+1})$ implies (conditional) exchangeability of

$$|Y_{n_0+1} - \hat g_{n_0}(\mathbf{X}_{n_0+1})|, \dots, |Y_n - \hat g_{n_0}(\mathbf{X}_n)|, |Y_{n+1} - \hat g_{n_0}(\mathbf{X}_{n+1})|$$

Split conformal predictor[10]

The data: with $n_0 + n_1 = n$

$$\underbrace{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_{n_0}, Y_{n_0})}_{\text{training} \rightsquigarrow \hat g_{n_0}}, \underbrace{(\mathbf{X}_{n_0+1}, Y_{n_0+1}), \dots, (\mathbf{X}_n, Y_n)}_{\text{calibration} \rightsquigarrow \hat q_{n_1}^{(1-\alpha)^+}}, \underbrace{(\mathbf{X}_{n+1}, Y_{n+1})}_{\text{test}}$$

- Quantile: $\hat q_{n_1}^{(1-\alpha)^+} = \hat Q_{n_1}^{(1-\alpha)(1+\frac{1}{n_1})}\left(\{|Y_i - \hat g_{n_0}(\mathbf{X}_i)|\}_{i=n_0+1}^n\right)$

Prediction intervals for test point $\mathbf{X}_{n+1}$,

$$\hat\Gamma(\mathbf{X}_{n+1}) = \left[\hat g_{n_0}(\mathbf{X}_{n+1}) \pm \hat q_{n_1}^{(1-\alpha)^+}\right]$$

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[10] Also known as *inductive conformal predictor* (Papadopoulos et al., 2002; Vovk et al., 2005, Algorithmic Learning in a Random World

Split conformal prediction

Theorem (*)

*Vovk et al 2005, Algorithmic Learning in a Random World

If $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_{n+1}, Y_{n+1})$ are exchangeable, then

$$\mathbb{P}\left(Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1})\right) \geq 1 - \alpha$$

Proof.
- Recall the notation: $S_i = |Y_i - \hat g_{n_0}(\mathbf{X}_i)|$ for $i = n_0 + 1, \dots, n, n+1$
- By construction

$$\begin{aligned} Y_{n+1} \in \hat\Gamma(\mathbf{X}_{n+1}) &\iff S_{n+1} \leq \hat q_{n_1}^{(1-\alpha)^+} = \hat Q_{n_1}^{(1-\alpha)(1+1/n_1)}\left(\{S_i\}_{n_0 < i \leq n}\right) \\ &\iff S_{n+1} \leq \hat q_{n_1+1}^{(1-\alpha)} = \hat Q_{n_1+1}^{(1-\alpha)}\left(\{S_i\}_{n_0 < i \leq n+1}\right) \end{aligned}$$

that holds with prob. $\geq 1 - \alpha$ ☐

Other splitting strategies

~~Cross-validation conformal prediction~~

~~Leave-one-out CV (also known as jackknife):~~

$$\hat\Gamma^{cv}(\mathbf{X}_{n+1}) = \left[\hat g(\mathbf{X}_{n+1}) \pm \hat Q_n^{1-\alpha}\left(\{|Y_i - \hat g_{-i}(\mathbf{X}_i)|\}_{i=1,\dots,n}\right)\right]$$

- ~~$\hat g_{-i}$ is built on $\mathcal{D}_n \setminus \{(\mathbf{X}_i, Y_i)\}$~~
- ~~Valid in-sample but may have zero coverage out-of-sample, in the worst case~~

~~More computationally efficient: K-fold CV:~~
- ~~Partition $\{1, \dots, n\} = A_1 \cup \dots \cup A_K$, with $|A_k| = n/K$~~
- ~~Fit models $\hat g_{-A_k}$ to data $\{(\mathbf{X}_i, Y_i)\}_{i \notin A_k}$~~
- ~~Compute non-conformity scores~~

$$\{|Y_i - \hat g_{-A_k}(\mathbf{X}_i)|\}_{k=1,\dots,K; i \in A_k}$$

~~Jackknife+~~

~~Jackknife can equivalently be defined as:~~

$$\begin{aligned} \hat\Gamma^{cv}(\mathbf{X}_{n+1}) &= \left[\hat Q^\alpha(\hat g(\mathbf{X}_{n+1}) - S_i), \hat Q^{1-\alpha}(\hat g(\mathbf{X}_{n+1}) + S_i)\right] \\ &= \left[-\hat Q^{1-\alpha}(-\hat g(\mathbf{X}_{n+1}) + S_i), \hat Q^{1-\alpha}(\hat g(\mathbf{X}_{n+1}) + S_i)\right] \end{aligned}$$

~~A modified version of the method: the jackknife+[11]~~

$$\hat\Gamma^{cv+}(\mathbf{X}_{n+1}) = \left[-\hat Q^{(1-\alpha)(1+1/n)}(-\hat g_{-i}(\mathbf{X}_{n+1}) + S_i), \hat Q^{(1-\alpha)(1+1/n)}(\hat g_{-i}(\mathbf{X}_{n+1}) + S_i)\right]$$

~~**Theorem (coverage for jackknife+)**~~

~~*If $Z_1, \dots, Z_{n+1}$ are exchangeable, and $\mathcal{A}$ is symmetric, then jackknife+ satisfies*~~

$$\mathbb{P}\{Y_{n+1} \in \Gamma(\mathbf{X}_{n+1})\} \geq 1 - 2\alpha$$

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[11] Barber et al. 2019, Predictive inference with the jackknife+

Other conformity scores

Conformal predictors relies on conformity measures
- Regression
  - $\rightsquigarrow s(\mathbf{x}, y) = |\hat g(\mathbf{x}) - y|$ (homoscedastic)
  - $\rightsquigarrow s(\mathbf{x}, y) = \frac{|\hat g(\mathbf{x}) - y|}{\hat\sigma(\mathbf{x})}$ (heteroscedastic)
  - $\rightsquigarrow s(\mathbf{x}, y) = \max\{y - \hat g_{\tau_-} ; \hat g_{\tau_+} - y\}$ (robustness)
    with $\hat g_\tau = \arg\min_g \frac{1}{n}\sum_{i=1}^n \rho_\tau(Y_i - g(\mathbf{X}_i))$ and $\rho_\tau(u) = (\tau - \mathbf{1}(\tau \leq 0))u$
  - ... so many other choices!
- Classification
  - $\rightsquigarrow s(\mathbf{x}, y) = \arg\min_{g \in \mathcal{G}} \ell(y, g(\mathbf{x}))$
- Many ~~other problems (supervised or not!)~~ $\rightsquigarrow$ LLM

Length considerations: optimal solutions

What about width?

- By construction (split) conformal predictor is valid
- Different score functions lead to different widths
- Different algorithms lead to different widths

Why width does not appear in the definition!

$$\hat\Gamma \in \arg\min_\Gamma \{\mathbb{E}[|\Gamma(\mathbf{X})|]\} \quad \text{s.t. } \mathbb{P}(Y \in \Gamma(\mathbf{X})) \geq 1 - \alpha\}$$

Width of split conformal predictor

In the case of regression[12]

Controlling the length of $\hat\Gamma(\mathbf{X})$ requires more conditions:
- $\mu(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] < \infty$ for all $\mathbf{x} \in \mathcal{X}$.
- $\epsilon = Y - \mu(\mathbf{X}) \perp\!\!\!\perp \mathbf{X}$ and density function of ϵ is symmetric and nonincreasing on $[0, \infty)$.
- (stability). There exists some $\tilde\mu$ such that for large enough n

$$\mathbb{P}\left(\|\hat g_n - \tilde\mu\|_\infty \geq \eta_n\right) \leq \rho_n, \quad \eta_n, \rho_n \to 0$$

and density of $|Y - \tilde\mu(\mathbf{X})|$ is bounded away from 0

Then

$$\hat q_n - q_n = \mathcal{O}(\eta_n + \rho_n + n^{-1/2})$$

where q_α is the population α quantile of $|Y - \hat g(\mathbf{X})|$

Distribution-free does not hold anymore!!

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[12] Lei *et al.*, Distribution-Free Predictive Inference For Regression, 2017

So let's make assumptions!

Could we solve optimally

$$\Gamma^* \in \arg\min_\Gamma \{\mathbb{E}[|\Gamma(\mathbf{X})|]\} \quad \text{s.t. } \mathbb{P}(Y \in \Gamma(\mathbf{X})) \geq 1 - \alpha\} ?$$

Under mild conditions:
- Y given $\mathbf{X}$ has density $p(\cdot|\mathbf{X})$
- For all $y \in \mathbb{R}$, the mapping $t \mapsto \mathbb{P}_{\mathbf{X}}(p(y|\boldsymbol{X}) \geq t)$ is continuous on $\mathbb{R}_+^*$.

$$\Gamma^*(\mathbf{X}) = \{y : p(y|\mathbf{X}) \geq t_\alpha^*\}$$

where $t_\alpha^* = H^{-1}(1-\alpha)$, with $H(t) = \mathbb{E}\left[\mathbf{1}_{\{p(Y|\mathbf{X}) \geq t\}}\right]$.

Estimation strategy:
- sample data $(\mathbf{X}_i, Y_i)$ to estimate $p(y|\mathbf{x})$
- sample data $(\mathbf{X}_i, Y_i)$ to estimate $H(t)$ and then t_α^*

Why don't we control the length instead?

$$\Gamma_\ell^* \in \arg\min\{\mathbb{P}(Y \notin \Gamma(\boldsymbol{X})), \quad \text{s.t. } \mathbb{E}[L(\Gamma(\boldsymbol{X})] \leq \ell\}$$

Under mild conditions:
- For all $y \in \mathbb{R}$, the mapping $t \mapsto \mathbb{P}_{\mathbf{X}}(p(y|\boldsymbol{X}) \geq t)$ is continuous on $\mathbb{R}_+^*$.

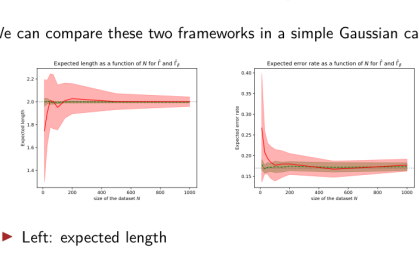
$$\Gamma_\ell^*(\boldsymbol{X}) = \{y \in \mathbb{R} : p(y|\boldsymbol{X}) \geq \lambda_\ell^*\}$$

where $\lambda_\ell^* = G^{-1}(\ell)$ with $G(t) := \int_{\mathbb{R}} \mathbb{P}(p(y|\boldsymbol{X}) \geq t) dy$

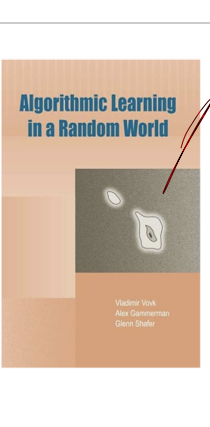
Estimation strategy:
- sample data $(\mathbf{X}_i, Y_i)$ to estimate $p(y|\mathbf{x})$
- sample *unlabeled* data $\mathbf{X}_i$ to estimate $G(t)$ and then λ_ℓ^*

Small numerical comparison

We can compare these two frameworks in a simple Gaussian case



- Left: expected length
- Right: expected error
- Dotted line: true parameters
- Dashed: $\hat\Gamma$
- Solid red: $\hat\Gamma_\alpha$



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Candès → popular